

SHAMIK BASU

Los Angeles, CA | +1 (213) 255-6219 | shamik1900@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Scientist and MS Data Science graduate at USC (GPA 3.73) with 2+ years of production experience building RAG pipelines, LLM inference workflows, and ML-driven analytics systems at scale. Designed systems processing 5M+ records/month, cut operational costs by 72%, and won 1st Place at UCLA SAIRS Hackathon 2025. Passionate about solving novel, cross-industry analytical problems.

EDUCATION

University of Southern California - MS, Data Science | GPA: 3.73 / 4.0 Jan 2025 – Dec 2026

Coursework: Machine Learning, Deep Learning, Data Management, Statistical Inference

SRM Institute of Science & Technology - BTech, Computer Science | GPA: 3.46 / 4.0 May 2018 – Jun 2022

Coursework: Algorithms, Artificial Intelligence, Probability & Queueing Theory, Data Structures

WORK EXPERIENCE

Data Science Associate Intern | KCC Capital Partners | Los Angeles, CA Jan 2026 – Present

- Fine-tuning an open-source SLM into a production JavaScript/Docker chatbot to automate client service request classification and routing, reducing manual handling overhead for the automation team.

Data Scientist | Bajaj Finserv Health | Pune, India Nov 2023 – Dec 2024

- Architected a real-time medical document analytics system using RAG, LangChain, and REST APIs processing 5M+ records/month at 92% extraction accuracy, replacing an entirely manual workflow.
- Engineered LLM-based inference pipelines (GPT-3.5 Turbo) to automate high-complexity decision workflows, cutting operational costs by 72% and built observability pipelines with Langfuse reducing GPU utilization by 15%.
- Integrated ML outputs into Power BI dashboards, cutting ad-hoc reporting TAT by 42% and enabling self-serve analytics for senior stakeholders.

Associate Data Scientist | Bajaj Finserv Health | Pune, India July 2022 – October 2023

- Developed supervised ML model for workforce performance prediction delivering statistically-validated insights improving efficiency by 22%; redesigned NER fraud detection pipeline increasing policyholder identification accuracy by 27%.
- Processed 10M+ records in Azure Synapse (SQL) for BI reporting; integrated Salesforce lead allocation framework using geographic proximity modeling, reducing customer churn by 33%.

Data Engineer Intern | Bajaj Finserv Health | Pune, India Jan 2022 – June 2022

- Ran A/B tests and cohort analyses improving web conversion by 37%; designed distributed analytics system in C++, Trino, and Docker supporting 10M+ records across 200+ features.

PROJECTS

Agentic Trend Orchestrator | <https://trendpilot.shamik-basu.com/>: May 2025 – Present

- Designed and deployed a full-stack agentic AI platform (FastAPI + Next.js + Docker) that automates trend discovery, predicts content performance via semantic NLP analysis, and orchestrates creator team workflows through a persistent multi-agent memory layer.
- Engineered session-guarded auth, real-time group chat, milestone-based project boards, and LLM-driven task generation from natural language notes - a generalizable agentic framework deployable across media, retail, or any trend-sensitive vertical.

EcoMateAI | <https://ecomateai.streamlit.app/> April 2025

UCLA SAIRS Hackathon - 1st Place, Sustainability Track | Judged by panelists from Microsoft, IBM, NVIDIA, Google & LinkedIn

- Built Streamlit analytics dashboards powered by Gemini 2.5 Flash to model daily carbon footprints, surface personalized reduction recommendations, and gamify sustainability goal-tracking for end users.

LEADERSHIP

GRIDS Club, USC - Vice President | 250+ members, largest data science org at USC September 2025 – Present

USC Viterbi School of Engineering - Graduate Student Mentor | Mentoring 6 graduate students January 2025 – Present

McKinsey Forward Program | Selected Participant | 2026 Present

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, CUDA, Bash, R, JavaScript

Analytics: Predictive Modeling, A/B Testing, Statistical Inference, Cohort Analysis, KPI Design, Scenario Analysis

ML / AI: Logistic Regression, Classification, Clustering, NLP (NER, BERT, spaCy), Deep Learning, LLMs, RAG, Agentic AI, LangChain, Langfuse, scikit-learn, TensorFlow, PyTorch

Data Platforms: Pandas, NumPy, Azure Synapse, MySQL, MongoDB, PostgreSQL, Snowflake, Trino, Apache Kafka

Engineering: FastAPI, Next.js, Docker, Kubernetes, CI/CD, REST APIs, Power BI, Tableau, Streamlit